Few-Shot Hyperspectral Image Classification Using Meta Learning and Regularized Finetuning

Wenmei Li, *Member, IEEE*, Qing Liu, Yu Zhang, Yu Wang, *Member, IEEE*, Yuan Yuan, Yan Jia, and Yuhong He

***Abstract*— The use of deep learning (DL)-based hyperspectral image (HSI) classification has been made remarkable progress in recent years. However, obtaining sufficient labeled samples for training DL models remains a challenge. Transfer learning is effective in addressing the problem of HSI classification with limited labeled samples. However, cross-domain HSI classification using transfer learning remain difficult, as differences in ground object categories between two datasets make it challenging to transfer and learn accurate. To address this issue, we propose a simple yet effective method for HSI classification using model-agnostic meta-learning (MAML) and Regularized Fine-tuning (MRFSL). Our method uses optimized 3-D convolutional neural networks (3D-CNNs) model, aided by MAML and cutout data augmentation to enable cross-domain transfer learning and carry out the HSI classification with limited target samples. Experiments conducted on three HSI datasets demonstrate that the MRFSL method achieves excellent results compared to existing methods. Specifically, the overall accuracy (OA) of our proposed MRFSL method reached 91.81%, 71.04%, and 88.35%, when only five labeled samples for each category were randomly extracted from the Salinas, Indian Pines (IPs), and University of Pavia (UP) datasets, respectively.**

***Index Terms*— Cross-domain, few-shot learning (FSL), hyperspectral image (HSI) classification, meta-learning, regularized finetuning.**

I. Introduction

REMOTE sensing image, especially hyperspectral images (HSIs) offering abundant spectral band and information, is crucial for surface information acquisition [1], [2], [3], [4], [5], [6]. HSIs have found wide applications in various fields, including agriculture management [7] and environmental protection [8]. Their effective features detection capabilities make them particularly valuable in these domains. However, processing HSIs is time-consuming and labor-intensive due to their high-dimension information. Additionally, HSI classification requires the identification of a unique surface type of ground objects in each pixel. Therefore, a significant part of the HSI classification involves reducing time consumption and improving classification efficiency. Numerous researchers have made great progress on this topic with successful outcomes.

In the past decade, machine learning algorithms such as support vector machine (SVM) [9] and random forest [10] has proven their effectiveness in HSI classification. But achieving high efficiency and precision in HSI classification is challenging due to the strong correlation between bands. Thus, accurately distinguishing and classifying features based on these correlated bands becomes a complex task. Deep learning (DL) methods such as generative adversarial network (GAN), graph convolutional networks (GCNs) [12], 1-D convolutional neural networks (CNNs) [11], and 3D-CNNs [13] have been introduced to improve information utilization and yield better classification results. However, the performance of these DL-based methods relies heavily on sufficient labeled samples, which are difficult to obtain in practical applications. Therefore, recent research has focused on developing new approaches that utilize insufficient samples to achieve high HSI classification accuracy.

A number of DL-based methods have been proposed for HSI classification with limited labeled samples. Liu et al. [14] proposed a semisupervised CNNs which is capable of extracting deep features from HSIs with limited samples. Besides, some researchers attempt to introduce additional irrelevant datasets to enhance the model's feature learning ability. Rao et al. [15] introduced the relational network into the few-shot (FS) HSI classification to obtain spatial and spectral similarity information between different samples. Li et al. [16] proposed a two-branch CNN-based framework for FS HSI classification that leverages the feature information extracted from self-supervised branch to enhance the classification performance of FS branch. Li et al. [17] designed a cross-domain HSI classification framework based on domain adaptation methods, and it utilizes the information from the source domain dataset to solve the problem of insufficient target domain samples in FS HSI classification. Recently, Yao et al. [18] proposed

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a framework based on semisupervised learning and active learning to generate pseudolabel for the unlabeled samples. This is also an important idea for HSI FS classification. However, many FS classification methods often suffer from overfitting and require improved generalization for better model performance. There are significant differences in ground object types between different HSI datasets, and how to achieve cross-domain feature adaptation in the model has become a key issue. Meta learning provides a good solution to this problem.

Meta-learning has effectively addressed the FS over-fitting problem by enabling machines to "learn to learn," and it has shown excellent performance in HSI classification with few samples. Meanwhile, regularization has appeared explicitly in HSI classification [19], and data augmentation is an classic regularization method; it is also effective for HSI classification as it expand new data from the original dataset to enhance the variety of the samples. Liu et al. [20] proposed a framework based on prototype network that significantly improved the accuracy of HSI FS classification. Further, a classical optimization-based meta-learning method, model-agnostic meta-learning (MAML) algorithm, achieves remarkable performance by training an appropriate initialization parameter for the model, and updating the hyperparameters through a few gradient steps. The trained model can perform well in new tasks, and the gradient updating step is calculated by using a few data from the new task. Gao et al. [21] introduced the MAML algorithm into the FS HSI classification and obtained satisfactory results.

However, the existing studies have limitations due to using a relatively small number of land cover classes in the source and target datasets, which do not fully exploit the advantages of the MAML algorithm. Additionally, insufficient test datasets limits the evaluation of MAML effectiveness in FS classification with HSIs. Besides, many existing FS HSI classification methods suffer from overfitting, which seriously affects the effectiveness of the model. To address these concerns, we proposed a method called MAML and regularized fine-tuning FS learning (MRFSL) for FS HSI classification. In this framework, MAML strategy and cutout data augmentation are introduced to optimize the training model and classify new tasks well through several gradient updates, while augmenting limited samples with the partially obscured version of the existing samples.

The main contributions of this article are highlighted as follows.

1) We present a hybrid FS learning (FSL) approach for achieving cross-domain adaptation in HSI classification. This approach combines the utilization of MAML and regularized fine-tuning.
2) By employing MAML in both the source and target domains, our method reduces the reliance on the source domain and facilitates the discovery of deep transferable knowledge.
3) Regularization methods are incorporated into the fine-tuning process to enhance the generalization of the model. Ablation experiments are conducted to evaluate the functionality of each module within this model.

II. Related Works

Previously, HSI classification predominantly relied on dimensionality reduction algorithms and machine learning-based classifiers. Dimensionality reduction techniques like principal component analysis (PCA) [22], auto-encoder (AE) [23], and linear discriminant analysis (LDA) [24] were utilized to reduce the dimensionality of HSIs, thereby enhancing performance and efficiency. Concurrently, machine learning-based classifiers such as SVM, extreme learning machine, and random forest were commonly employed for classification. Recently, there has been a surge in the introduction of novel dimensionality reduction methods [25], [26] and DL-based classification algorithms, aimed to further elevating classification efficiency and accuracy.

A. DL-Based HSI Classification Methods

According to the feature types applied, DL-based HSI classification methods can be divided into three categories, the spatial features-based methods, the spectral features-based methods, and the spectral-spatial features-based methods [27].

1) Spatial Features-Based Methods: Spatial features-based methods leverages geometric attributes of land covers, including their location, shape, size, and spatial relationship with adjacent objects. Yue et al. [28] first introduced a DCNNs-LR framework that utilizes convolution layers and logistic regression classifier for HSI classification, achieving superior performance over other machine learning methods. To account for the high spatial correlation of HSIs, a pixel-centered window can be used to define the neighborhood of a pixel when extracting spatial features, since most adjacent pixels belong to the same category [29]. Additionally, DBNs can be used to extract spatial information by considering pixels as image patches, which are originally regarded as spectral vectors [30].

2) Spectral Feature-Based Methods: Spectral features are the unique spectral characteristics of different land covers at different electromagnetic waves. HSIs contain rich spectral information that distinguishes them from other types of images, and therefore, spectral features are very important in HSI classification. Chen et al. [31] first used the DL methods based on stacked autoencoder (SAE) for HSI classification and found that SAEs methods outperformed traditional machine learning algorithms by efficiently extracting deep spectral features.

Further, Yu et al. [32] designed a CNN structure and the 1×1 convolutional kernels and dropout layers to address overfitting issue. Compared with the unsupervised SAE method, the supervised CNN method not only extracted more accurate spectral information but also enhanced the generalization of the model. More recently, Pei et al. [33] designed a dual-channel spectral enhancement network (DSEN) that uses two 3-D–2-D hybrid convolution channels to reduce network parameters and achieve excellent classification results.

3) Spectral-Spatial Features-Based Methods: Spectral-spatial features refer to the fusion of spectral and spatial features extracted from HSIs. Lin et al. [34] proposed a DL methods that combines autoencoder and PCA to extract

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spectral-spatial features, which outperformed classical classifiers and offered a novel approach for HSI classification. Song et al. [35] proposed a framework based on residual learning (RL) and 3-D GAN to extract more spectral-spatial features from HSIs. The framework uses RL to address the negative impact of increasing the number of layers in DL models and extracts the deep spectral-spatial features using a fusion network. Guo et al. [36] proposed a model with spectral-spatial connected attention mechanism (FG-SSCA) for HSI classification. Specifically, the FG-SSCA method first extracts the coarse spectral-spatial features using the spectral and spatial attention mechanism, and then further extracts deep features using the residual module. Experiments demonstrated that this framework has great superiority and generalization ability in HSI classification.

However, the above-mentioned HSI classification methods usually require adequate samples for the model. If there are not sufficient samples for training, the mentioned methods may overfit. Thus, some researchers have focused on the FS HSI classification and attempted to apply FSL algorithms to reduce the training samples required for the above schemes.

B. FS HSI Classification Methods

1) Metric Learning-Based Methods: The essential idea of metric learning is "learn to compare." Different from other classification methods, metric learning is to obtain a more meaningful or separable feature space by learning the similarity between data. In recent years, the metric-based FSL has been introduced into HSI classification. Relation networks is a classic method of metric learning. Rao et al. [15] proposed a relation network based FSL method for HSI classification. They used a relation network to compare the similarity of the spatial-spectral features that extracted from the images. Xi et al. [37] proposed a FSL framework for HSI classification with a class-covariance metric. This method uses Mahalanobis distance measurement for label discrimination during testing and constructs more accurate decision boundaries than conventional metrics. Recently, Dong et al. [38] proposed a clustered metric learning method for FS HSI classification, which combines manifold learning and metric learning. This approach constructs pixels around the sample into a manifold unit, replacing the samples for distance comparisons, and enhances the effect of metric learning using the idea of manifold learning.

2) Meta Learning-Based Methods: The essential idea of meta learning is "learn to learn," and it is a promising research direction of FSL. The concept of meta learning originates from educational psychology and there are three main categories of meta learning methods: metric-based, model-based, and optimization-based [39].

In recent years, researchers have successfully introduced meta-learning method into FS HSI classification. Liu et al. [20] first introduced meta learning method for HSI classification. Specifically, Liu et al. [20] designed a framework that combines prototype network and RL to improve the performance of FS HSI classification. Their results demonstrated that the meta learning-based methods outperformed traditional FSL methods. Gao et al. [40] introduced metric-based learning using the relation network into HSI classification. They further proposed an optimization-based method based on MAML for FS HSI classification, where a simple CNN model was applied to validate the classification effect of the MAML method with limited samples. In addition, they also explored the performance of MAML in cross-domain and cross-data classification [21]. Although existing meta-learning methods applied to HSI classification have achieved remarkable performance, the development of meta-learning based HSI classification method is still rare. Moreover, due to the lack of samples in the target datasets, most existing meta learning methods still suffer from overfitting. Thus, further research in this area is required to explore more novel and effective meta-learning methods to improve the accuracy and efficiency of HSI classification and reduce overfitting of the model.

3) Data Augmentation-Based Methods: Data augmentation techniques involve generating new training samples from the original dataset. It is an effective way to address the issue of limited labeled samples and prevent overfitting. Yu et al. [41] introduced three data augmentation methods, namely flip, translation, and rotation for FS HSI classification. These methods were applied to expand the training data and achieved improved results in HSI classification. Nalepa et al. [42] proposed a data augmentation that generates new samples at test time, which enhances the generalization of deep models. Specifically, instead of creating new samples before the model training, the method generates new examples during the inference. This method not only improves the classification accuracy of deep neural network, but also improves its generalization ability. Dou et al. [43] proposed a novel data augmentation method based on HSI denoising, which is specifically designed for HSI and maintains good classification performance on HSIs with the presence of Gaussian noise. Data augmentation is an effective regularization method, but using data augmentation alone often cannot achieve good classification results.

4) Hybrid Methods: Hybrid method involve combining several algorithms or techniques into a framework. As the field of HSI classification continues to evolve, many algorithms and techniques have been proposed to improve classification accuracy. Consequently, fusion models that combine multiple algorithms has gained significant attention. For instance, Bai et al. [44] proposed a hybrid network that combines RL, attention mechanism, meta learning, and feature transformation. The RL and attention mechanism were utilized to extract features, while the feature transformation strategy was applied to diversify the features. The results show that the proposed framework outperformed existing FS classification methods. Another hybrid method was designed by Li et al. [45] which combines semisupervised graph convolutional network (SSGCN) with metric learning. The classification accuracy of this framework outperformed previously proposed methods, while requiring less computation. The results demonstrate the potential of hybrid methods for improving the classification accuracy of HSI classification.

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III. Problem Description

A. HSI Classification Problem

The HSI classification is to assign each pixel of the HSI to the corresponding category. The input sample of the method is represented by x, which is a HSI pixel cube collected by the hyperspectral imager. The HSI dataset can be represented by $\mathcal{D} = \{\mathbf{x}_n, \mathbf{y}_n\}_{n=1}^{N}$. The N denotes the entire dataset number. The training and validation datasets can be divided into $\mathcal{D}^{\text{tr}}$ and $\mathcal{D}^{\text{va}}$

$$\mathcal{D}^{\text{tr}} = \{(\mathbf{x}_n, \mathbf{y}_n)\}_{n=1}^{N_t} \tag{1}$$

where N_t denotes the training samples number. The HSI classification is to learn a mapping function f from $\mathbf{x}$ to $\mathbf{y}$ with parameters θ

$$\hat{\mathbf{y}} = f(\mathbf{x}; \theta) \quad \text{s.t. } (\mathbf{x}, \mathbf{y}) \in \mathcal{D}^{\text{tr}} \tag{2}$$

$$\theta^* = \arg\min_{\theta} \sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}^{\text{tr}}} \mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) \tag{3}$$

where $\hat{\mathbf{y}}$ denotes the predicted category of the samples, and the optimal parameters obtained through training are then validated for accuracy on the validation set. $\mathcal{L}(\cdot)$ can measure the difference between predicted and true values.

It is known that a generalizable DL model can be easily obtained with sufficient labeled samples, but difficult to achieve when labeled sample is insufficient, which is the problem of FSL is proposed to address.

B. FS HSI Classification Problem

The challenge is to train a well-generalized DL model with limited samples. The FS HSI classification can be described as C-way K-shot problem, where C and K denote the number of the category and samples per category. In general, the value of the K is small, such as 1, 5, and 10. The training set of FS HSI classification can be defined as $\mathcal{D}^{\text{fs}} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^{C\times K}$. And the testing set is defined as $\mathcal{D}^{\text{te}} = \{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=1}^{N_{\text{te}}}$, which has sufficient samples of the same category as D^{fs}. The traditional HSI classification methods according to formula (2) and (3) do not achieve satisfactory classification results on the small number of sample dataset $\mathcal{D}^{\text{fs}}$ and suffer from overfitting phenomenon. To address this issue, the training paradigm of pretraining and fine-tuning is a common solution. Hence, in addition to the fine-tuning dataset, this method requires the introduction of a pretraining dataset $\mathcal{D}^{\text{pt}}$. The $\mathcal{D}^{\text{pt}}$ contains sufficient samples of HSIs with their corresponding labels. If the categories in dataset $\mathcal{D}^{\text{pt}}$ differs from that in dataset $\mathcal{D}^{\text{fs}}$, this leads to a cross-domain FS HSI classification problem. Where, $\mathcal{D}^{\text{pt}}$ is the source dataset and the samples of $\mathcal{D}^{\text{fs}}$ are extracted from the target dataset. The purpose of the source dataset is to provide additional transferable feature information for the model.

The FSL model aims to learn the tasks by constantly updating the parameter θ. The initial parameters of the model are defined as θ_0, and the model parameter update process of the pretraining process is as follows:

$$\theta_{\text{pt}} = \arg\min_{\theta} \sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}^{\text{pt}}} \mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) \tag{4}$$

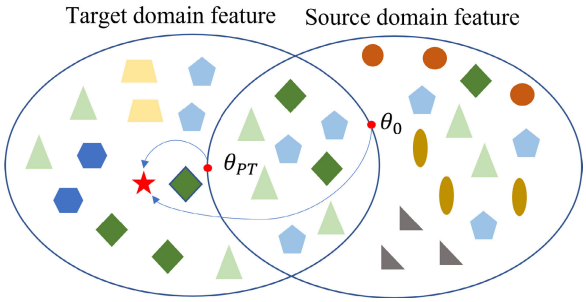


Fig. 1. Optimization process of the Pretrain-Fine-tuning training paradigm. "↶" represents the optimization process and the red "★" represents the optimization objective.

$$\theta_{\text{pt}} \leftarrow \theta_0 - \eta_1 \frac{\partial \mathcal{L}}{\partial \theta} \tag{5}$$

where η_{pt} represents the learning rate in the pretraining process. Then, the model parameter θ_{pt} is fine-tuned with the samples of $\mathcal{D}^{\text{fs}}$. And the process of the fine-tuning is shown as follows:

$$\theta_{\text{ft}} = \arg\min_{\theta} \sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}^{\text{fs}}} \mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) \tag{6}$$

$$\theta_{\text{ft}} \leftarrow \theta_{\text{pt}} - \eta_{\text{ft}} \frac{\partial \mathcal{L}}{\partial \theta} \tag{7}$$

where, η_{ft} represents the learning rate in the fine-tuning process. Finally, the fine-tuned model parameter θ_{ft} is obtained. The optimization process is shown in Fig. 1. As shown in the figure, the pretraining of the model can speed up the optimization process and enables faster and more accurate convergence of the model parameters. Specifically, the model can extract some transferable information from the pretraining process to reduce the number of fine-tuning gradient updates and obtain higher classification performance. In sum, the FSL method is generally achieved by introducing additional learnable information to improve the classification ability and generalization of models when there are insufficient labeled samples. This concept is feasible and can improve model performance because the introduction of additional knowledge is easy to understand and implement.

IV. Proposed Approach

Meta learning, as a type of FSL, has been proven to be a highly effective FSL method. However, due to the small number of samples in the target domain, many FS classification methods often suffer from overfitting, and require improved generalization for better model performance. Therefore, the combination of meta learning and regularization methods can effectively achieve high-precision FS HSI classification. The proposed method is a CNN-based model that combined with meta-learning and regularization methods, which is expected to tackle cross-domain HSI classification problems with limited labeled samples. Given its demonstrated performance of information extraction, CNN is adopted to obtain feature information of HSIs. MAML is the meta learning method applied to train the initialization parameters in the pretraining

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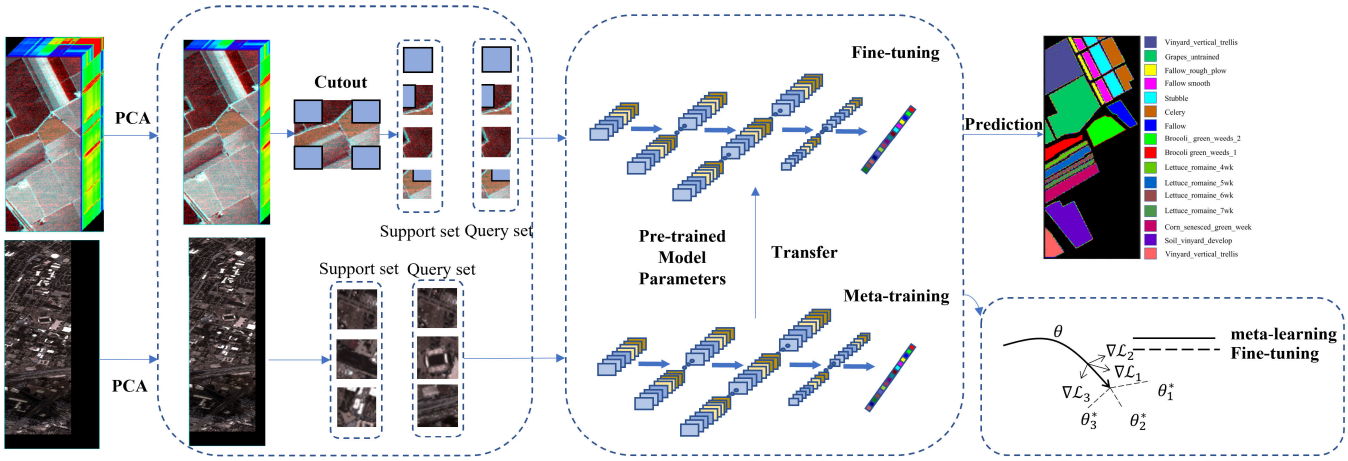


Fig. 2. Architecture of the proposed framework.

process. Regularized fine-tuning is to use the data augmentation method to further improve robustness and generalizability of the network.

A. Framework of MRFSL

The MRFSL framework for FS HSI classification follows the pretraining, fine-tuning, and testing paradigm commonly used in FSL methods.

The system architecture of MRFSL is presented in Fig. 2, consisting of a pretraining MAML-CNN based model on the source dataset, and regularized fine-tuning on the target dataset, and its algorithm is described clearly in Algorithm 1. The proposed method aims to address the cross-domain hyperspectral classification problem, and thus the source and target domain are completely different datasets with different categories and pixel numbers.

Pretraining is realized by a CNN model, and MAML strategy is also applied during the pretraining. Fine-tuning is achieved by the MAML strategy and regularization methods. At last, the fine-tuned model combined with the test dataset produced in target domain are used for classification.

B. Pretraining With MAML Strategy

In the pretraining process, the basic model is a 3D-CNN model. Indeed, it is difficult to utilize a basic model to obtain model parameters that perform well on new tasks with limited samples. To enable the basic model to achieve better classification performance, the source domain HSI dataset D^{pt} was introduced to learn the transferable features. In order to learn more transferable knowledge, the optimization process of the model needs to be improved. Thus, we introduced the MAML algorithm in the process of pretraining.

The core of MAML is to train the initialization parameters of the model. The model is trained by episodes, which are usually described as C-way K-shot N-query problems, and they can simulate the FS classification tasks. There are two sets in an episode, support set and query set. The support set contains $C \times K$ labeled samples and the query set contains $C \times N$ unlabeled samples. Where the C denotes the class number, and the K and N denotes the sample number selected from D^{pt}. When faced with new tasks, the model updates the parameters on the support set, but calculates the objective loss on the query set.

The MAML algorithm aims to learn the model parameters θ_{pt} that exhibit a high degree of sensitivity to variations in the task of the source domain dataset, such that slightly changes in the parameters, such as the direction of the gradient, will lead to large improvements on the model. The $p(\mathcal{T})$ represents the tasks selected from the D^{pt}. Formally, f_θ represents the model and the θ represents parameters in the model. When working with a new task $\mathcal{T}_i$, the parameter of the model θ will be replaced by θ'. According to formula (5), one gradient update of the model can be expressed as formula (8). In MRFSL method, we used multiple gradient updates to make the model fully learned

$$\theta_i' \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta) \tag{8}$$

where α is the inner learning rate. By optimizing the performance of $f_{\theta'}$ with respect to θ to learn model parameters. Where the task was selected from $p(\mathcal{T})$. Specifically, the objective is illustrated in the following equation:

$$\min_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'}) = \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}\left(f_{\theta - a\nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)}\right). \tag{9}$$

While the model parameters θ is optimized by Adam as expressed in the following equation:

$$\theta_{\mathrm{pt}} \leftarrow \theta - \beta \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}\left(f_{\theta_i'}\right) \tag{10}$$

where β is the meta learning rate. It is noted that model parameters θ is applied to make meta-optimization and the updated model parameters θ' is utilized to compute the objective [46].

The updating of the MAML gradient is achieved by the structure of a gradient through a gradient. Specifically, the original model is updated by calculating the gradient using the sum of the losses of each task from query set. And the loss of the query set is calculated using model parameters derived from tasks selected on the support set.

Through the two learning rates and multiple gradient updates, the base model can learn more general feature information. After pretraining, the θ_{pt} that are sensitive to the new task in the source domain are obtained. According to the principles of MAML, the pretrained model parameters are also sensitive to new tasks in other domains.

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C. Regularized Fine-Tuning

We utilized the principle of vicinal risk minimization (VRM) in the fine-tuning process to prevent overfitting. The fine-tuning process was based on the target domain dataset $\mathcal{D}_{\text{ta}}$

$$\mathcal{D}_{\text{ta}} = \left\{\mathcal{D}^{\text{fs}}, \mathcal{D}^{\text{te}}\right\} \tag{11}$$

where $\mathcal{D}^{\text{fs}}$ and $\mathcal{D}^{\text{te}}$ represent the fine-tuning and testing dataset. Similarly, there are also two different sets in the fine tuning dataset.

The pretrained model in the traditional FSL cross-domain classification method is fine-tuned directly on dataset D^{fs}, and since the number of samples generally selected for fine-tuning is small in the FS problem, there is still a possibility of overfitting in the fine-tuning process, and the model may also suffer from poor performance. Regularization is an effective solution.

The principle of machine learning is to minimize the expected average of the loss function over the data distribution

$$\min_{\theta^* \in \theta} \varepsilon_{\text{em}} = \min_{\theta^* \in \theta} \int \mathcal{L}\left(\hat{\mathbf{y}}, \mathbf{y}\right) \mathrm{d}P(\mathbf{x}, \mathbf{y}) \tag{12}$$

where the θ and $P(\mathbf{x}, \mathbf{y})$ represent the weight space and the joint distribution. θ^* can minimize empirical error. However, $P(\mathbf{x}, \mathbf{y})$ is usually unknown in formula (12). Therefore, according to empirical risk minimization theory, $P(\mathbf{x}, \mathbf{y})$ can be approximately replaced by empirical distribution

$$P_\delta(\mathbf{x}, \mathbf{y}) = \frac{1}{C \times K} \sum_{t=1}^{C \times K} \delta(\mathbf{x} = \mathbf{x}_t, \mathbf{y} = \mathbf{y}_t) \tag{13}$$

where the C_2 denotes the class number in D^{fs} and the K_2 denotes the samples number in support set selected from D^{fs}. The objectives of the machine learning model combined with the regularization method can be represented as

$$\min_{\theta^* \in \theta} \varepsilon_{\text{re}} = \min_{\theta^* \in \theta} \varepsilon_{\text{em}} + \lambda_{\text{re}} R\left(\theta^*\right). \tag{14}$$

$R(\cdot)$ denotes the regularization term. λ_{re} denotes the regularization term coefficient, which balances the empirical error with the regularization term. Common regularization methods include l_1/l_2 regularization, dropout, and early stopping. These methods are based on the principle of structural risk minimization. Besides, the regularization methods based on the principle of minimizing neighborhood risk also perform well. The VRM principle is to approximate the joint distribution using a vicinal distribution

$$P_v\left(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}\right) = \frac{1}{C \times K} \sum_{t=1}^{C \times K} \nu\left(\tilde{\mathbf{x}}, \tilde{\mathbf{y}} \,\middle|\, \mathbf{x}_i, \mathbf{y}_i\right). \tag{15}$$

$\nu(\cdot)$ represents the probability of finding the virtual sample $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ in the vicinity of the real sample$(\mathbf{x}_i, \mathbf{y}_i)$. According to VRM principles, $R(\cdot)$ is computed by

$$\begin{aligned} R\left(\theta^*\right) = & \sum_{m=1}^{M} \mathcal{L}\left[f\left(\tilde{\mathbf{x}}_m; \theta^*\right), \tilde{\mathbf{y}}_m\right] \delta\left(\tilde{\mathbf{x}}_m, \tilde{\mathbf{y}}_m\right) \\ & \text{s.t.} \left(\tilde{\mathbf{x}}_m \neq \mathbf{x}_n, \tilde{\mathbf{y}}_m \neq \mathbf{y}_n\right) \quad \forall \left(\mathbf{x_n}, \mathbf{y_n}\right) \in \mathcal{D}^{\text{fs}} \end{aligned} \tag{16}$$

where M represents the augmented samples number.

1) Cutout Data Augmentation: Data augmentation has been widely used in practice when training CNNs. Cutout is one of the classic data augmentation methods, and it is a regularization skill for CNNs that involves randomly removing contiguous sections of input images, effectively augmenting the dataset with partially occluded images of existing samples [47]. In order to determine the erasing range, the vertex coordinates $E = (c_x, c_y, c_w, c_h)$ of the erasing area boundary need to be determined first. The aspect ratio of the erasing area is proportional to the original image

$$\begin{aligned} c_x &\sim \text{Unif}(0, W), \quad c_w = W\sqrt{1-\lambda} \\ c_y &\sim \text{Unif}(0, H), \quad c_h = H\sqrt{1-\lambda}. \end{aligned} \tag{17}$$

The λ is calculated as $(c_w c_h / WH) + 1$, W denotes the image width and H denotes the image height. The value of erasing area is filled with 0.

This method can be regarded as an extension of dropout layer in neural networks, and it can be used in data preprocessing. To implement cutout, some fixed-size masks are applied to a random location of each input HSI cube during each epoch of training. The area erased by the cutout method is not necessarily in the original image. If it is at the edge of the HSI cube, the erased area will not be a square. If the model can still classify correctly after inputting a part of the occluded image into the model, then the model will certainly force the network to use the local data that is not occluded for recognition, which will increase the training difficulty and generalization. Cutout is also a method to add noise to images to increase the robustness of the model. The size and shape of the cutout region are important hyperparameters. We conducted our experiments using a square patch as the cutout region to remove consecutive sections of the input HSI cubes. It is well-known that the spectral data tend to suffer from various degradation, noise effects, or variabilities in the process of imaging [48]. By increasing the difficulty of model training, this method enhances the model's antiinterference ability to a certain extent.

V. Experiments and Discussion

The experiments are carried out on a PC with Intel Xeon Silver 4210R @ 2.40 GHz CPU and GTX 3070 graphical processing unit (GPU). The open-source software framework used in this article is Pytorch.

A. Experimental Datasets

1) Source Domain Dataset: The Houston data are commonly used dataset for HSI classification. The scene was acquired in 2012, and it covers the University of Houston campus and its neighborhoods. There are 349 × 1905 labeled pixels and 15 land-cover classes in the Houston dataset (see Fig. 3). The spatial resolution is 2.5 m by pixel and the band number is 144. In addition, the Indian Pines (IP) dataset was also selected to verify the impact of the source dataset on the model. The IP dataset is described in Section V-A2.

2) Target Domain Dataset: The Salinas dataset is also one of the benchmark datasets, which encompasses the region of Salinas Valley in CA, USA. The dataset comprises

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Algorithm 1 MRFSL Method for FS HSI Classification

Input: $p(\mathcal{T})$, α, β, Source domain D^{pt}, and Target domain D^{ta} **Output:** Weights θ_{ft}

[Parameter initialization]:
M: the augmented samples number; N_2: the number of samples in query set selected from the Target domain.
[Pretraining with MAML Strategy]:
randomly initialize θ
while not done **do**
  Sample batch of tasks $\mathcal{T}_i \sim p(\mathcal{T})$ from D^{pt}
  for all $\mathcal{T}_i$ **do**
    Select $C \times K$ samples $\mathcal{D} = \{x^{(i)}, y^{(i)}\}$ from $\mathcal{T}_i$
    Evaluate $\nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$ using D and $\mathcal{L}_{\mathcal{T}_i}$
    Compute adapted parameters with gradient descent: $\theta_i' = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$
    Select $C \times N$ samples$\mathcal{D}_i' = \{x^{(j)}, y^{(j)}\}$ from $\mathcal{T}_i$ for the meta-update
  end for
  Update $\theta_{pt} \leftarrow \theta - \beta \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta_i'})$ using $\mathcal{D}_i'$ and $\mathcal{L}_{\mathcal{T}_i}$
end while
[Regularized Fine-tuning]:
for $t = 0, 1, 2, \ldots, C \times (K + N) - 1$ from D^{fs} **do**
  Select the erasing area according to (17) and obtain its augmented samples $\mathcal{D}^a = \{\tilde{\mathbf{x}}_m, \tilde{\mathbf{y}}_m\}_{m=1}^{M}$
end
for $m = 0, 1, 2, \ldots, M - 1$ **do**
  Optimize the model parameters θ_{pt} according to the above process (5)-(14) according to the augmented samples and obtain the model parameters θ_{ft}
end

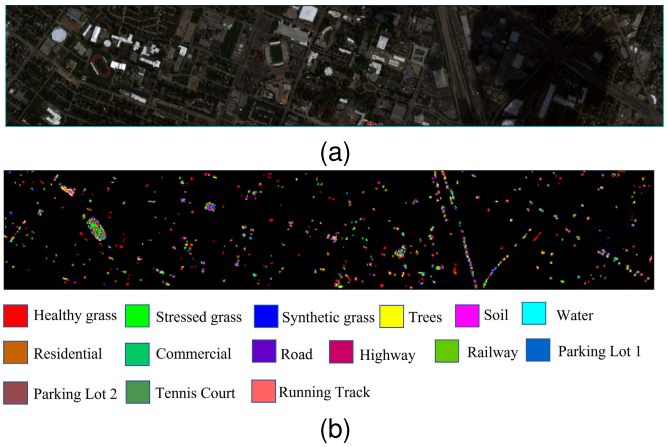


Fig. 3. Houston dataset. (a) False-color image. (b) Ground-truth map.

111 104 pixels, each measuring 2.5 m in spatial resolution [see Fig. 4(a)]. It contains 54 129 labeled pixels and 16 land-cover categories. The detailed information is shown in Fig. 4(b).

The IP is the first public dataset for HSI classification aided by DL. The dataset was captured in 1992 over the pine trees and their neighboring areas in Indiana, USA. It comprises 21 025 pixels, each measuring 20 m in spatial resolution. It contains 16 categories of land covers, mainly including Corn-notill, Alfalfa, and Soybean-mintill. The detailed information is shown in Fig. 5, Fig. 5(a) is the IP dataset in false-color format, and Fig. 5(b) illustrates its ground truth.



Fig. 4. Salinas dataset. (a) False-color image. (b) Ground-truth map.

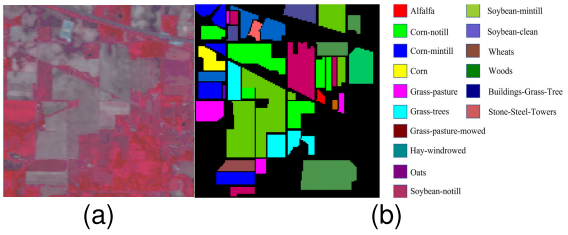


Fig. 5. IP dataset. (a) False-color image. (b) Ground-truth map.

TABLE I
DETAIL INFORMATION OF THE SOURCE AND TARGET DATASETS

l	Houston	IP	Salinas	UP
Pixels	349×1905	145×145	512×217	610×340
Spectral range	380-1050	400-2500	400-2500	430-860
band	144	200	204	103
GSD	2.5	20	3.7	1.3
Sensor type	ITRES-CASI 1500	AVIRIS	AVIRIS	ROSIS
Aera	Houston	Indiana	California	Pavia
No. of Categories	15	16	16	9

The University of Pavia (UP) dataset was captured over the area of the UP, Italy. It comprises 207 400 pixels, each measuring 1.3 m in spatial resolution. It contains nine categories of land covers, mainly including Asphalt, Meadows, and Gravel. The detailed information is shown in Fig. 6, and more information of the four mentioned datasets are presented in Table I.

B. Experimental Setting

During the experiments, numpy, sklearn, random, PIL, and spectral libraries are used. The network of the MCFSL method is an arbitrary DL network. In our experiments, a simple 3D-CNN model is designed, and its detail structure is shown in Fig. 7. The 3D-CNN model can effectively utilize deep spectral-spatial information of the HSIs. The Relu activation

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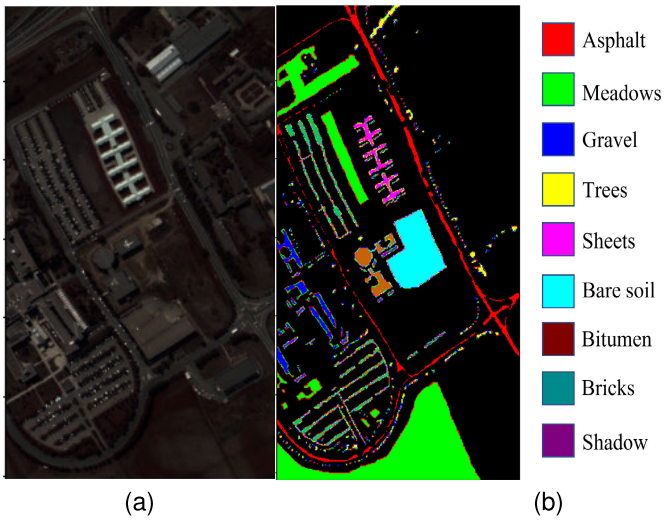


Fig. 6. UP dataset. (a) False-color image. (b) Ground-truth map.

function layer prevents the gradient from disappearing during model training. Meanwhile, the BN layer prevents the model from being over-fit. Finally, linear and softmax layers are used to implement the classification.

Due to the large amount of HSIs, analysis and processing based on these images requires a significant amount of computation. Moreover, data redundancy can also negatively impact HSI classification results. To mitigate the issue, the PCA is utilized to reduce the data redundancy of HSIs. The input to the model is the cubes around the target pixels with a size of $11 \times 11 \times N$ (N denotes the band number following PCA).

We attempt to analyze the impact of pretraining dataset on the proposed method. There are two different HSI datasets selected as the source dataset. In the experiments, during the meta-training, the support sets and query sets number were set to 10 and 20, respectively. Our results indicate that the MRFSL method exhibits varying performance on different datasets. Specifically, we selected the Salinas and IP datasets as the target domain datasets and used five and ten labeled target domain samples per class for fine-tuning. Additionally, to ensure more reliable and consistent results, we conducted ten independent repeated experiments for each experiment to minimize the impact of random sampling on the model's accuracy.

If a lot of edge samples of HSIs are selected, the extracted feature information will be insufficient, and thus we scramble the dataset and conduct random sampling. The episode number is set to be 200. There are two learning rates in the optimization process of the MRFSL method: the inner learning rate α and the meta learning rate β. These two learning rates control the ability of model optimization. Specifically, the α and β are set to 0.1 and 0.0001.

The commonly used evaluation metrics in HSI classification include overall accuracy (OA), accuracy of per class, kappa coefficient, and average accuracy (AA).

C. Ablation Study

To verify whether the MAML algorithm and the cutout module have corresponding effect on the classification performance. We removed the MAML and cutout module from the model both together and separately to explore the effect of each module, and the experiment results are presented in Table II. The C-3DCNN is the base model with Cutout data augmentation technique. In the MAML method, we removed the cutout data augmentation technique of the MRFSL method. Specifically, removing the MAML algorithm leads to a decrease of 4.42% in the OA of the model, which demonstrates that the MAML algorithm can greatly improve the performance of the base model and the cutout module can really improve the classification accuracy. We also explored the impact of the target domain labeled samples number on the MRFSL method. The classification performance of MRFSL increases with the increasing target domain labeled samples. The OA of MRFSL method in the case of ten labeled samples is increased by 1.26% compared to that of five labeled samples.

TABLE II
ABLATION EXPERIMENTS

Methods	shot5			shot10		
	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
MRFSL	91.89	93.29	0.9099	93.15	94.78	0.9238
MAML	91.66	93.49	0.9072	92.65	93.81	0.9182
C-3DCNN	87.47	87.71	0.8611	91.35	92.48	0.9039
3DCNN	82.11	78.68	0.8028	88.17	88.68	0.8687

D. Compared With Other Methods

We selected the following machine learning, DL, and FSL techniques for the purpose of comparison in the HSI classification.

1) Support Vector Machine: SVM is a widely used machine learning method in HSI classification. It has advantages in addressing classification problems with limited samples. But, the ability of this method to solve multiclass problems is insufficient.

2) 2D-CNNs and 3D-CNNs: Both the 2D-CNNs and 3D-CNNs are classic supervised DL networks. They could be used to acquire spectral and spatial features, and then process them to classify the HSIs. 2D-CNNs and 3D-CNNs have become commonly used effective methods in HSI classification.

3) DCFSL: DCFSL is a new proposed framework for tackling FSL and domain adaptation issues. The conditional adversarial domain adaptation strategy and FSL method are used to overcome domain shift and learn transferable knowledge in the source domain. This method has a better performance compared with the regular FSL methods in HSI classification.

4) RNFSL: RNFSL is a metric-based meta learning method [40]. It utilizes feature learning modules and relationship learning modules to extract deep features of HSIs.

5) S3NET: S3NET is a recently proposed method based on Siamese networks [49]. Siamese network is also a classic meta learning method. The base model of the S3NET is a lightweight spectral–spatial network (SSN) composed of 1-D and 2-D convolution. Then, the Siamese networks are used to enhance the effectiveness of the model.

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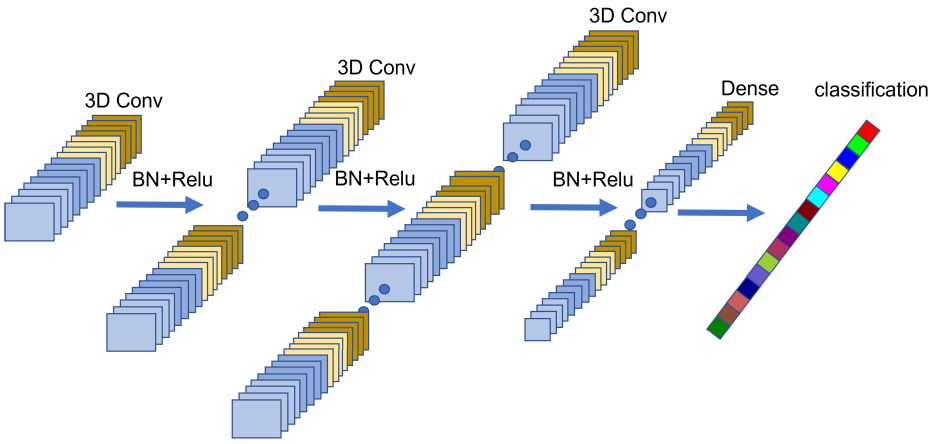


Fig. 7. Structure of the CNN method.

TABLE III
CLASSIFICATION ACCURACY ON THE SALINAS DATASET (FIVE LABELED SAMPLES PER CATEGORY)

Class	SVM	2D-CNNs	3D-CNNs	DCFSL	MAML	RNFSL	S3NET	MRFSL
1	97.19	18.62	97.87	98.24	99.81	99.21	99.72	**99.93**
2	91.96	90.81	99.08	99.33	99.57	99.08	95.54	**99.65**
3	76.75	43.17	96.39	91.75	98.76	97.01	**99.01**	98.48
4	98.24	96.22	98.41	98.59	99.25	94.19	98.05	**99.25**
5	92.92	48.10	81.75	95.01	96.85	**99.88**	97.33	97.37
6	97.73	97.38	98.08	**100.00**	99.70	98.59	99.71	99.75
7	98.47	74.92	**99.73**	98.12	99.53	95.25	99.72	99.42
8	56.72	43.34	70.98	59.65	82.21	**86.15**	77.10	81.77
9	94.16	65.94	95.47	99.88	**99.91**	94.64	98.76	99.84
10	67.44	72.31	67.25	89.63	94.90	**96.96**	92.35	94.83
11	90.39	67.14	96.84	98.63	98.80	98.34	98.96	**99.35**
12	87.96	**99.38**	88.78	99.13	98.59	92.93	93.08	98.19
13	97.52	83.95	90.03	**98.59**	77.68	96.66	94.51	97.16
14	88.61	93.45	96.91	**99.71**	99.03	96.14	93.44	98.86
15	60.00	58.65	61.35	73.05	71.92	74.84	**84.01**	73.63
16	83.72	66.66	92.76	**99.81**	97.81	93.80	94.87	97.63
OA(%)	79.38±2.33	61.16±10.54	83.20±4.29	86.41±0.86	91.34±0.52	91.33±0.83	91.24±1.66	**91.81±0.24**
AA(%)	86.26±2.18	64.56±15.36	83.72±4.36	93.69±0.42	92.26±2.30	94.6±0.64	**94.76±1.26**	93.57±0.37
Kappa(×100)	77.11±2.57	57.34±11.29	81.42±4.78	84.94±0.95	90.37±0.58	90.36±0.92	90.27 ±1.83	**90.89±0.26**

TABLE IV
CLASSIFICATION ACCURACY ON THE IP DATASET (FIVE LABELED SAMPLES PER CATEGORY)

Class	SVM	2D-CNNs	3D-CNNs	DCFSL	MAML	RNFSL	S3NET	MRFSL
1	83.42	28.29	86.58	97.56	97.22	21.88	98.29	**100.00**
2	29.46	3.95	32.36	22.22	51.69	49.50	46.35	**54.41**
3	32.86	26.29	41.38	46.98	**61.59**	44.81	56.55	55.15
4	45.56	39.44	31.77	93.02	69.60	55.06	85.34	**95.59**
5	52.45	23.58	66.13	71.06	87.32	**93.19**	80.42	88.69
6	75.08	91.86	91.53	88.74	95.14	87.40	92.65	**97.12**
7	90.00	87.39	98.26	**100.00**	**100.00**	45.99	**100.00**	**100.00**
8	64.95	32.85	91.69	95.52	**100.00**	98.32	**100.00**	99.12
9	92.00	53.33	**100.00**	**100.00**	**100.00**	36.53	**100.00**	**100.00**
10	41.09	5.39	23.05	65.00	54.78	**69.74**	59.59	63.46
11	44.09	9.78	29.45	65.81	40.37	**81.54**	55.22	62.36
12	33.35	6.04	16.77	37.68	36.19	**65.05**	63.71	44.31
13	89.15	46.70	98.95	97.70	**100.00**	68.02	99.40	99.95
14	62.62	20.25	62.45	91.43	74.82	**97.26**	86.48	84.62
15	29.11	27.24	56.35	78.27	72.07	77.50	76.64	**92.23**
16	86.93	24.20	100.00	95.91	**100.00**	58.58	99.20	**100.00**
OA(%)	47.16±2.86	21.04±7.40	46.14±1.98	65.15±0.06	64.88±0.67	69.29±1.65	67.12±3.43	**71.04 ±0.68**
AA(%)	59.51±2.25	30.03±11.11	45.17±8.45	77.93±0.01	77.97±0.84	65.65±1.99	80.21±1.24	**83.56±0.43**
Kappa(×100)	41.00±2.68	15.44±6.91	40.92±2.18	60.71±0.08	60.66±0.71	65.93±1.78	63.22±3.56	**67.58±0.73**

To explore the impact of the target domain on MRFSL, we extracted the same number of labeled samples from Salinas, IP, and UP datasets for analysis. The detailed results are illustrated in Tables III–V. For the methods SVM, 2D-CNN, 3D-CNN, and S3NET, the number of samples selected was five. For the methods RNFSL, DCFSL, and MRFSL, only five

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TABLE V
CLASSIFICATION ACCURACY ON THE UP DATASET (FIVE LABELED SAMPLES PER CATEGORY)

Class	SVM	2D-CNNs	3D-CNNs	DCFSL	MAML	RNFSL	S3NET	MRFSL
1	66.98	52.50	45.76	81.65	69.13	90.30	67.49	**87.38**
2	59.68	56.36	52.08	89.06	84.98	**94.33**	73.76	87.57
3	55.07	52.59	86.92	65.21	82.92	65.22	70.32	**93.25**
4	89.50	86.01	65.33	**93.50**	82.59	88.93	92.40	87.56
5	98.99	98.65	99.91	85.05	99.27	99.03	99.41	**99.73**
6	47.15	56.27	75.08	87.72	94.18	70.24	70.05	**92.25**
7	83.96	92.39	97.31	86.49	**99.95**	64.92	96.09	98.83
8	68.51	72.02	35.28	62.36	**85.88**	76.50	74.16	76.16
9	99.81	99.19	70.03	92.63	**99.07**	95.31	98.38	98.23
OA(%)	64.88±4.32	62.42±6.26	58.31±6.06	78.53±3.89	84.63±0.61	85.91±1.08	75.59±6.79	**88.35±0.5**
AA(%)	74.41±1.40	74.00±3.44	69.74±2.99	83.24±2.27	88.66±0.38	82.75±1.44	82.45±2.89	**91.22±0.26**
Kappa(×100)	56.29±4.22	53.80±6.84	52.18±8.15	68.58±10.45	80.25±0.74	81.52±1.31	69.33±7.48	**84.90±0.61**

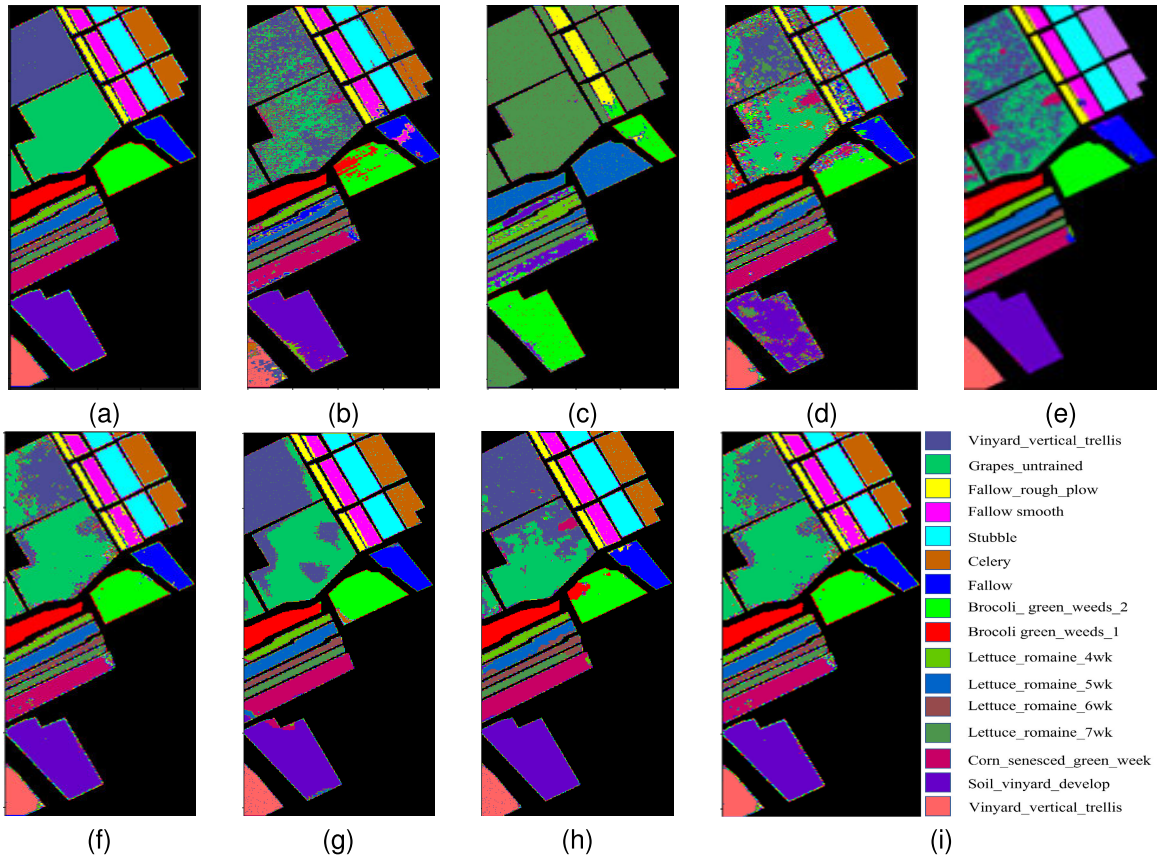


Fig. 8. Classification maps of the Salinas dataset (five labeled samples per category). (a) Ground truth. (b) SVM. (c) 2D-CNNs. (d) 3D-CNNs. (e) DCFSL. (f) MAML. (g) RMFSL. (h) S3NET. (i) MRFSL.

samples per category are extracted from the target domain to construct the fine-tuning datasets. To ensure the fairness of the experiment, in all the cross-domain method experiments, we selected the same source domain to learn transferable knowledge.

Experiments demonstrated that the MRFSL method clearly outperforms SVM, 2D-CNN, and 3D-CNN methods. Compared with SVM and 3D-CNN methods, MRFSL method increases OA by 12.43% and 8.61%, AA by 7.31% and 9.85%, kappa by 13.78% and 9.47% on Salinas dataset, respectively. However, compared with the newly proposed HSI FSL method DCFSL, RNFSL, and S3NET, the accuracy of MCFSL is greatly improved.

Besides, our proposed method also performs well on IP dataset. Compared with RNFSL, S3NET, and DCFSL methods, MRFSL method increases OA by 1.75%, 3.92% and 5.89%, AA by 17.91%, 3.35% and 5.59%, kappa by 1.84%, 4.36% and 6.87%, respectively. And the similar result can also be obtained on the classification results of the UP dataset. Compared with RNFSL, S3NET and DCFSL methods, MRFSL method increases OA by 2.44%, 12.76% and 9.82%, AA by 8.47%, 8.77% and 7.98%, kappa by 3.38%, 15.57% and 16.32%, respectively.

For the DCFSL method, the Chikusei dataset results in the best performance of the source domain. This study investigates the performance of the DCFSL method on two different

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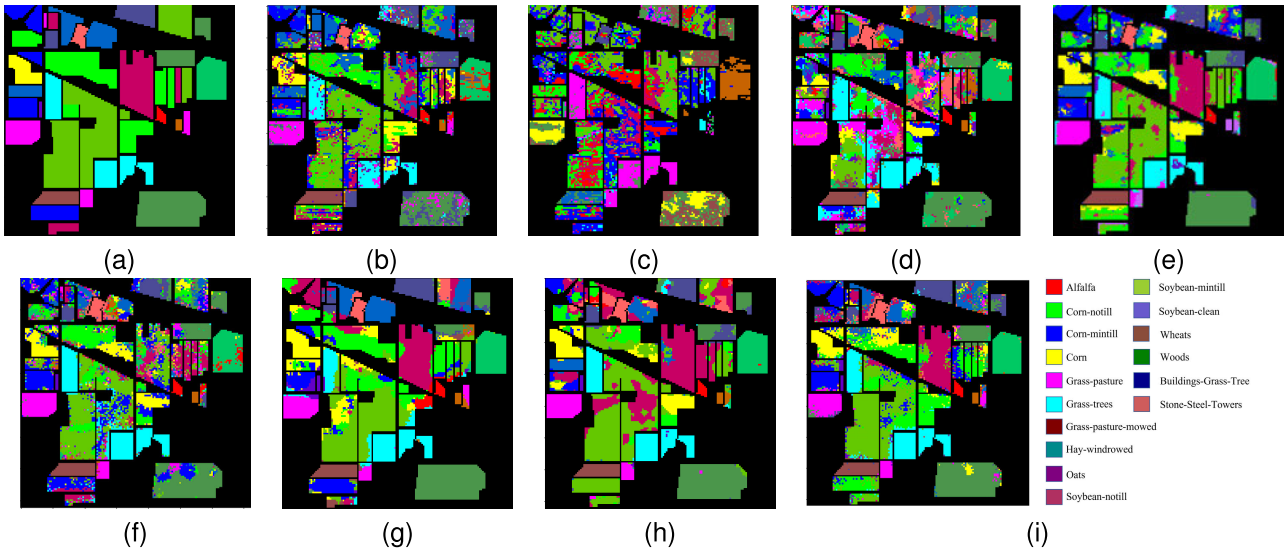


Fig. 9. Classification maps of the IP dataset (five labeled samples per category). (a) Ground truth. (b) SVM. (c) 2D-CNNs. (d) 3D-CNNs. (e) DCFSL. (f) MAML. (g) RMFSL. (h) S3NET. (i) MRFSL.

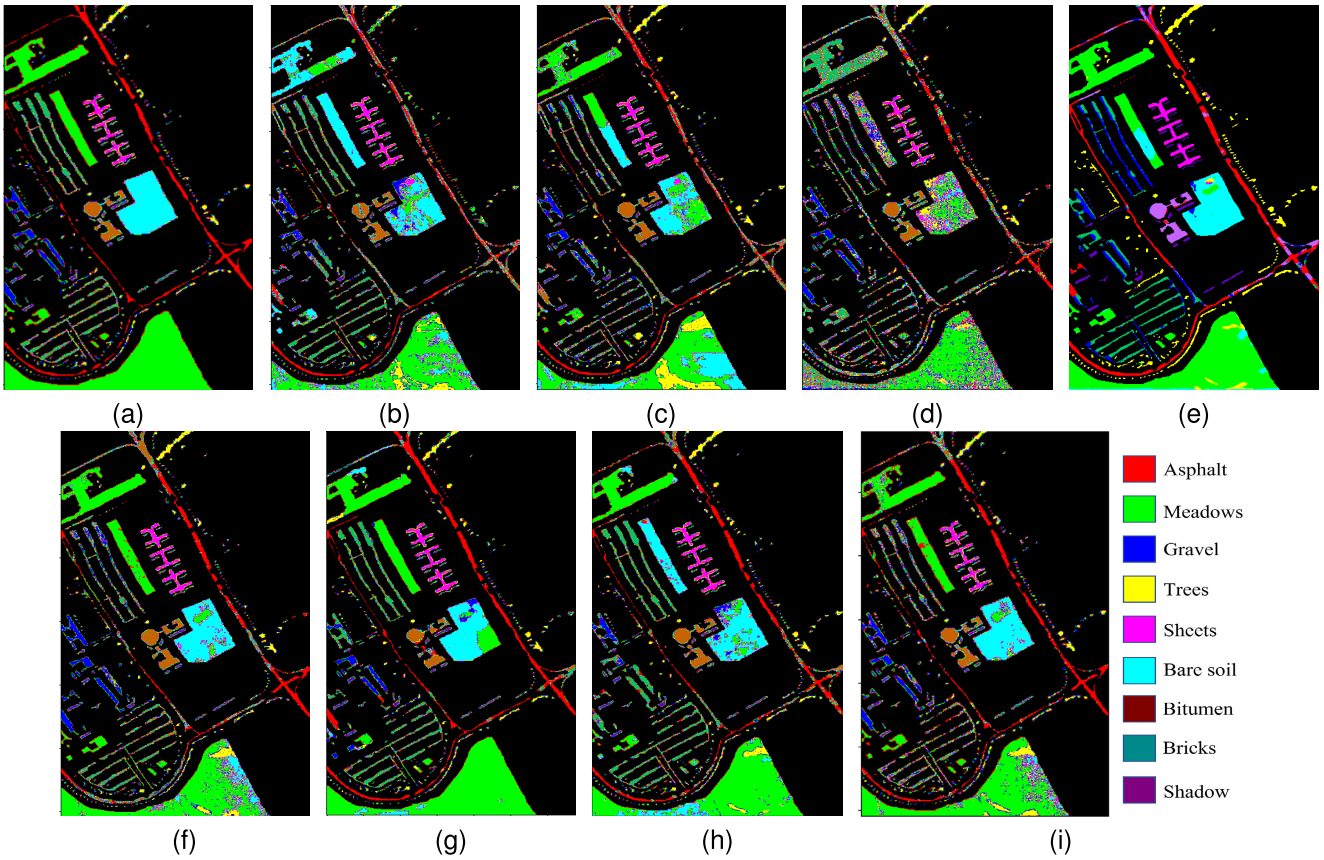


Fig. 10. Classification maps of the UP dataset (five labeled samples per category). (a) Ground truth. (b) SVM. (c) 2D-CNNs. (d) 3D-CNNs. (e) DCFSL. (f) MAML. (g) RMFSL. (h) S3NET. (i) MRFSL.

source datasets, namely the Chikusei dataset and the Houston dataset, and compares their respective results. The difference is marginal. Therefore, using the Houston dataset as the source domain for the DCFSL method will not significantly impact DCFSL performance.

Figs. 8–10 are the classification maps of the SA, IP and UP dataset with five labeled samples selected for training per class. Fig. 8 shows that the classification map of MRFSL is much better than the maps resulted from other methods. In addition, when only five labeled samples used in the training, the classification results obtained by 2D-CNN are the least ideal. The 3D-CNN model performs slightly better than SVM method. This indicates that the traditional supervised machine learning and DL methods cannot achieve excellent classification results with limited samples.

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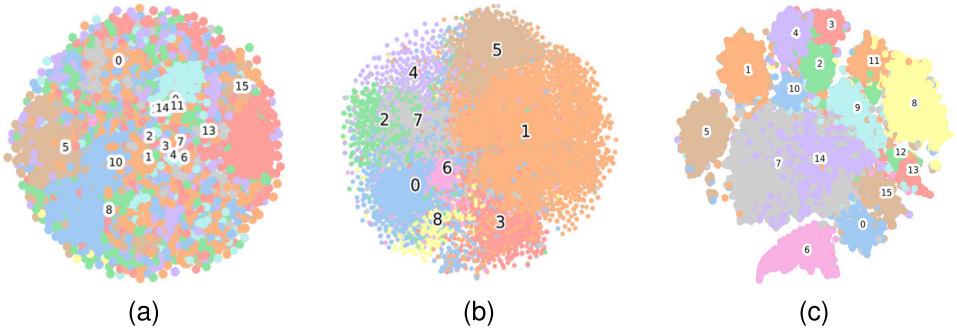


Fig. 11. Visualization of the embedding features of the MRFSL method in the three datasets [0–15 corresponds to categories 1–16 of the IP/UP/Salinas dataset (order of publicly available data categories)]. (a) IP. (b) UP. (c) Salinas.

The DCFSL method performs well when there are few samples in both the source and target domain. If the sample number selected from the source dataset increases, the model may obtain information that interferes with the classification. When we use the DCFSL method for comparative experiments, we modified the source domain dataset and corresponding parameters in the source code published by the authors [17]. Therefore, the classification accuracy obtained by using this method is slightly lower than that in their article. The potential reason for the decrease in precision may be that when the DCFSL method selects different datasets as the source domain, there are some skills in setting hyperparameter that are not described in the article. Even if the DCFSL method selects the samples from the best source domain, MRFSL method still yields better classification results when classifying the Salinas dataset. Specifically, the classification performance has been improved by 2.47% and 2.72% in OA and kappa. Similarly, the superiority of MRFSL is also manifested in the classification of IP and UP datasets.

As shown in Figs. 8–10, the MRFSL method can achieve better classification results. This is because meta learning enables the model to learn how to learn and improve its generalization. The regularization method enhances the model's ability to resist interference, leading to fewer errors and noise in the model's classification results. Especially in categories such as Asphalt, Gravel, Painted metal sheets, and Bare soil in the UP dataset, it has higher accuracy and less noise in the classification map.

In summary, the MRFSL method has the best classification accuracy on three target domain datasets. The MRFSL method has a better performance for classes that are easily misclassified by other classification methods, such as the class 2 in Salinas dataset, class 2, 4, 6, and 15 in IP dataset and the class 1, 3, and 6 in UP dataset. After analyzing the features (as shown in Fig. 11), it was found that Salinas dataset has the highest accuracy and the best feature separability. The visualization results of its features can be easily mistaken for the eighth and 15th types of features, leading to the lowest classification outcomes. Similar phenomena also occur on IP and UP datasets.

E. Influence of the Choice of Source Domain

To investigate the influence of the source dataset on the MRFSL, contrasting experiments on the IP dataset are carried out. We explored the impact of the source domain on the proposed method by comparing the classification results of the MAML and MRFSL methods in the IP and Houston source domains, where five samples per category of ground objects are selected in the target domain. Table VI shows the classification results from IP or Houston data to the Salinas set. As shown in the table, MCFSL (Houston) method performed much better than MRFSL (IP). Specifically, the MRFSL (Houston) method surpasses the MRFSL (IP) method by 3.89%(AA), 10.27%(OA), 4.3%(kappa) on Salinas target dataset. Compared to MRFSL method, the MAML is more susceptible to the influence of different source datasets. The choice of source domain dataset is thus very important for the MRFSL method. The Houston dataset contains 15 categories and 349 × 1905 pixels, which is larger than the IPs. Larger images can extract more transferable knowledge for subsequent fine-tuning. Therefore, a good source dataset is essential for cross-domain HSI classification.

TABLE VI
COMPARISON BETWEEN DIFFERENT SOURCE DOMAIN DATASET

Methods	IP			Houston		
Metric	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
MAML	79.41	67.45	0.7718	91.66	93.49	0.9072
MRFSL	88.76	83.55	0.8752	92.65	93.82	0.9182

TABLE VII
COMPARISON BETWEEN DIFFERENT REGULARIZATION METHODS (TEN-SHOT)

Methods	Salinas		
Metric	OA(%)	AA(%)	Kappa
Cutout	93.15	94.78	0.9238
Mixup	92.97	94.80	0.9217
Cutmix	92.78	94.54	0.8980

F. Influence of the Choice of Regularization Methods

In the regularized fine-tuning procedure, in addition to the cutout data augmentation method, we also chose other data augmentation methods for comparison experiments, e.g., mixup and cutmix data augmentation methods. Table VII is the contrasting results of different regularization methods. Experiments show that the cutout method has some advantages over the other two methods in the regularized fine-tuning process. Specifically, the MRFSL (cutout) method surpasses the MRFSL (cutmix) method by 0.37%, 0.24%, and 2.58% of

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AA, OA, and kappa on Salinas with ten labeled samples per class.

VI. Conclusion

Our novel cross-domain FS HSI classification method, MRFSL, utilizes the MAML algorithm and cutout augmentation to improve classification accuracy with limited samples. The MAML algorithm provides the optimization framework, while cutout augmentation is a regularization technique that further enhances accuracy and stability. Ablation experiments confirm the importance of each module in the proposed method. MRFSL was tested on two HSI datasets and outperformed the state-of-the-art method. Future work should explore the potential of other meta-learning methods to achieve high accuracy under cross-domain and limited sample conditions in HSI classification tasks. More specifically, in terms of source domain, multiple source domain features can be used to obtain more extensive features for cross-domain transfer learning. In addition, the application value of the proposed MRFSL model can be enhanced by extending it to practical application scenarios.

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